

Treasure Car Algorithm - Updated Version

This version keeps the 5-step lookahead and pruning framework, but adds a tail-stage decision rule and updates the scoring rule.

Core planning method

- Use 5-step lookahead to evaluate future treasure choices.
- Execute only the first chosen treasure, then re-plan from the new state.
- Candidate sequences are ranked mainly by score/time (CP value), with score and shorter time as tie-breakers.

Pruning logic

1. Time pruning: stop expanding a branch once its time already exceeds the remaining usable time.
2. Score upper-bound pruning: stop expanding a branch if even taking all remaining treasure scores still cannot beat the current best reference.
3. Dominance pruning: for equivalent partial states, discard states that are both slower and lower-scoring than an existing state.

New tail-stage rule

When the remaining time is less than 9 seconds, the planner no longer uses 5-step lookahead. Instead, it computes the required time to each remaining treasure and chooses the reachable treasure with the highest score.

New scoring rule

All treasure scores are now 3 times the original version. In code, this is implemented by multiplying the original Manhattan-distance-based score by 3.

Effect on result interpretation

The route is still generated dynamically by the algorithm at runtime. The update changes the final stage behavior and magnifies score differences between treasures, making the high-value tail choice more decisive.

Change	Meaning
Remaining time < 9 s	Switch to reachable-highest-score selection
Score multiplier	New score = original score x 3
Route style	Algorithm-generated at runtime, not hard-coded